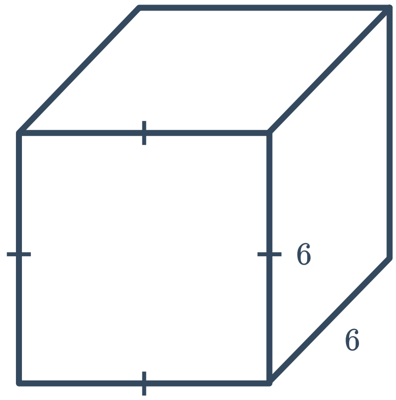
Surface area of prisms

1 Consider the following cube with a side length of $6\ \text{cm}$:



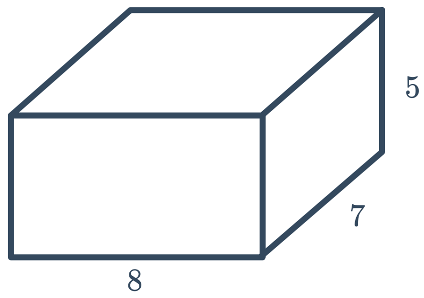
a Which of the following nets match the given cube?

A **B** **C** **D**

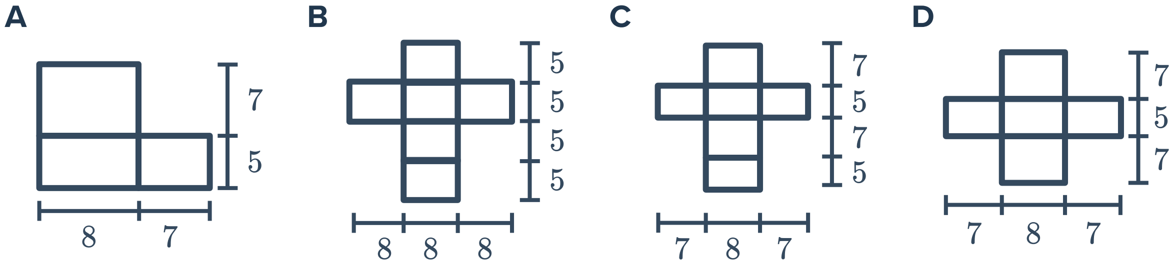
b Find the area of the cube's net.

c Find the surface area of the cube.

2 Consider the following rectangular prism:



a Which of the following nets match the given rectangular prism?

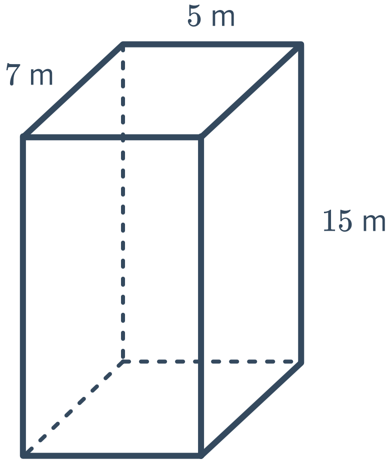


b Find the area of the prism's net.

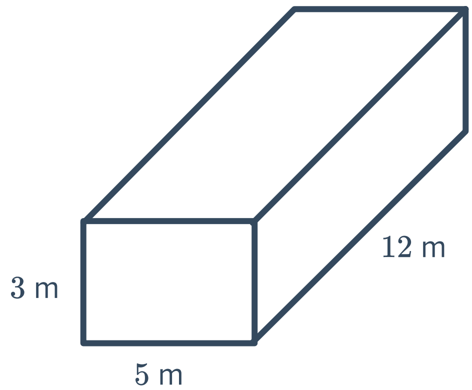
c Find the surface area of the rectangular prism.

3 Find the surface area of the following rectangular prisms:

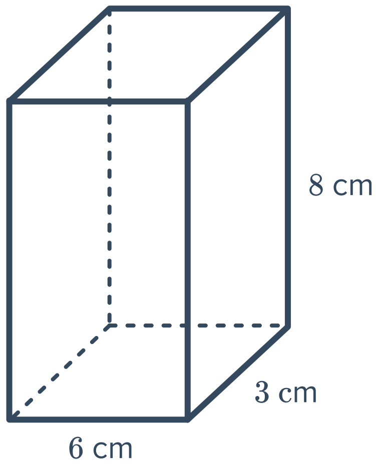
a



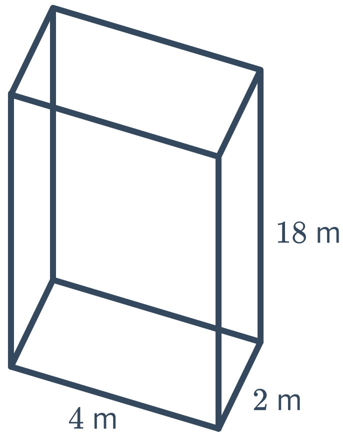
b



c

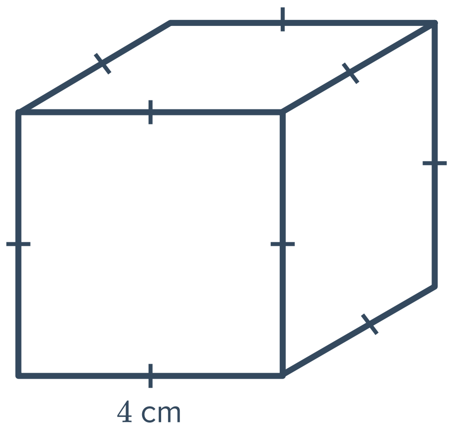


d

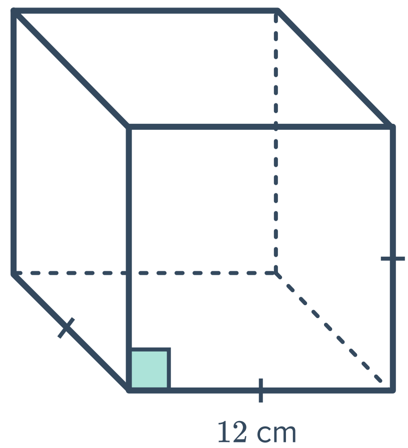


4 Find the surface area of the following cubes:

a



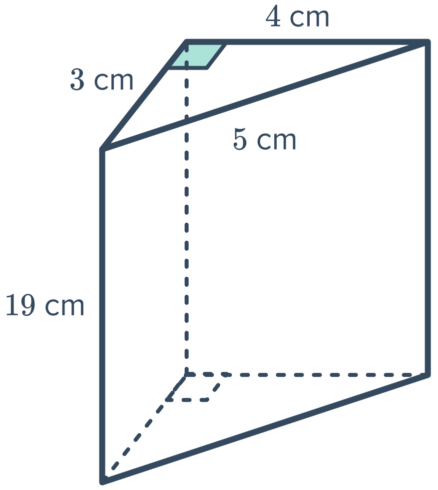
b



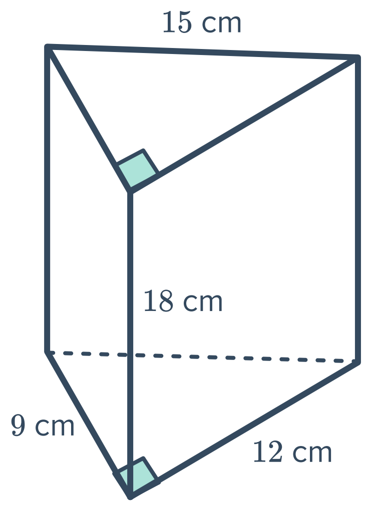
5 Find the side length of a cube that has a surface area of 384 cm^2.

6 Find the surface area of the following trian[redacted] prisms:

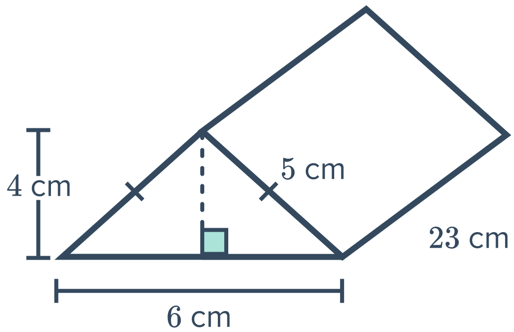
a



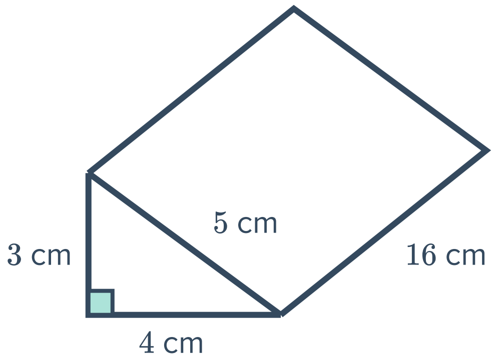
b



c



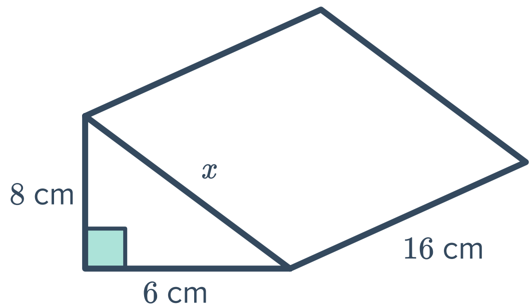
d



7 Consider the following figure:

a Find the value of x.

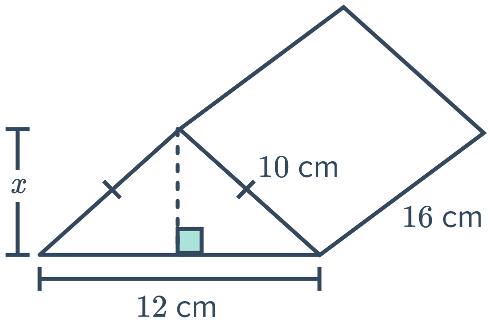
b Find its surface area.



8 Consider the following figure:

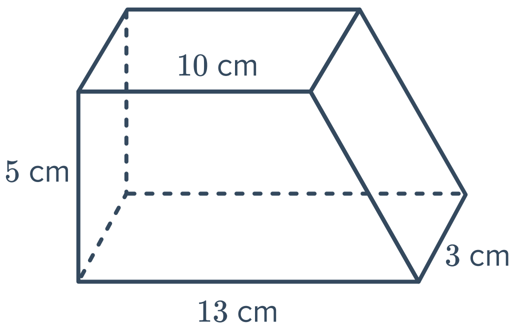
a Find the find the value of x.

b Find its surface area.

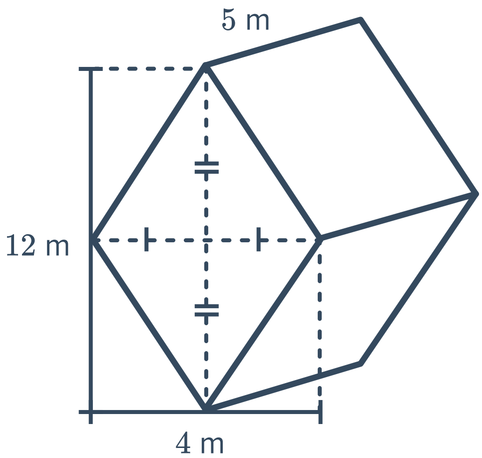


9 Find the surface area of the following prisms. Round your answer to two decimal places when necessary.

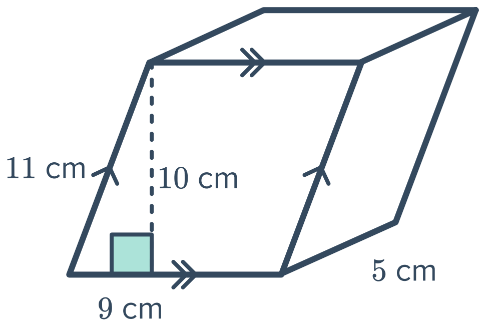
a



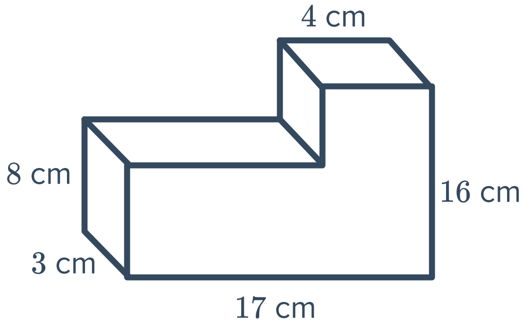
b



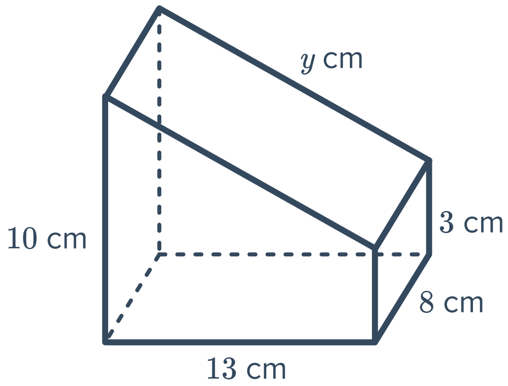
c



d



10 Consider the following trapezoidal prism:

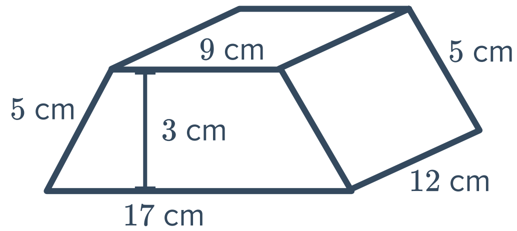


a) Find the value of y correct to one decimal place.

b) Find its surface area.

11 Consider the following trapezoidal prism:

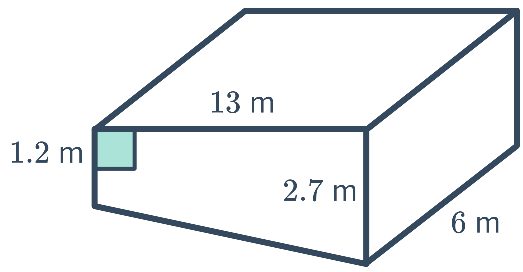
Find its surface area.



Applications

12 Consider the swimming pool with the following dimensions:

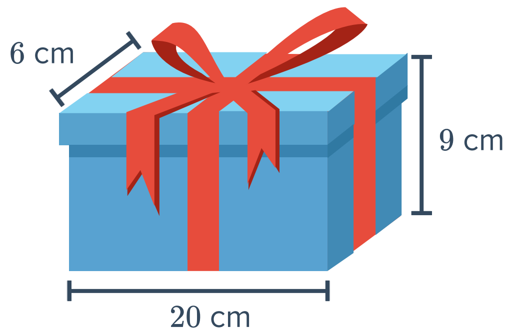
Calculate the area inside the pool that is to be tiled. Round your answer to two decimal places.



13 Sally is building a storage chest in the sha[redacted] a rectangular prism. The chest will be $93\ \mathrm{cm}$ long, $80\ \mathrm{cm}$ deep, and $18\ \mathrm{cm}$ high. Find the surface area of the chest.

14 A birthday gift is placed inside the box shown:

Find the minimum amount of wrapping paper needed to wrap this gift?



15 Both of the following popcorn bags are designed to carry $735\ \mathrm{cm}^3$ of popcorn.

a Find the height h of bag A.

b Find the height h of bag B.

c Find the surface area of bag A.

d Find the surface area of bag B.

e Which bag should be used to reduce the amount of paper needed?